

JEE MAIN REVISION MASTER: LINEAR PROGRAMMING

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Linear Programming — Core Principles & Governing Laws	8 PYQs	Core Principle
Moderate	Linear Programming — Standard Calculations & Problem Archetypes	7 PYQs	Quantitative Analysis
Standard	Linear Programming — Applied Cases, Reactions & Traps	2 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Linear Programming — Core Principles & Governing Laws	$\begin{aligned} a_1x_1 + a_2x_2 + \dots + a_nx_n &\leq b_i, \\ 2x - 3y + 6 &= 0 \quad \forall x, y \geq 0 \end{aligned} \quad \text{... (ii)} \quad \Rightarrow \frac{x}{4} + \frac{y}{5} = 1$
Linear Programming — Standard Calculations & Problem Archetypes	$z = px + qy, \quad 15p = 5q, \quad z = 4x + 6y$
Linear Programming — Applied Cases, Reactions & Traps	$z = px = qy$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Linear Programming — Core Principles & Governing Laws (High Yield — 8 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Linear Programming.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Linear Programming.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Linear Programming PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points ..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Consider the following statements: Statement (I): In a LPP, the objective function is always linear. Statement (II): In a LPP, the linear ine..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The shaded region in the figure given is the solution of which of the inequations?..." [Ans: D]

3.2 Linear Programming — Standard Calculations & Problem Archetypes (Moderate Yield — 7 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Linear Programming.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The corner points of the feasible region determined by the system of linear constraints are $(0,10), (5,5), (15,15), (0,20)$. Let $z=px+qy$ wh..." [Ans: D]
- [2025 Shift] Q#4 (MCQ): "The maximum value of $z=3x+4y$, subject to the constraints $x+y \leq 40, x+2y \leq 60$ and $x,y \geq 0$ is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "Corner points of the feasible region for an LPP are $(0,2), (3,0), (6,0), (6,8)$ and $(0,5)$. Let $Z=4x+6y$ be the objective function. The min..." [Ans: N/A]

3.3 Linear Programming — Applied Cases, Reactions & Traps (Standard Yield — 2 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Linear Programming.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2022 Shift] Q#4 (MCQ): "A dietician has to develop a special diet using two foods X and Y. Each packet (containing 30g) of food. X contains 12 units of ca..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "Corner points of the feasible region determined by the system of linear constraints are $(0,3), (1,1)$ and $(3,0)$. Let $z=px+qy$, where, $p, q > 0$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Linear Programming Problem (LPP), the objective function $Z=ax+by$ has the same maximum value at two corner points. The number of points at which $Z_{\text{(max)}}$ occurs is

- (A) 1
- (B) 2
- (C) 0
- (D) Infinity

Correct Answer: None

Solution Analysis: If the objective function $Z=ax+by$ has the same maximum value at two corner points of the feasible region, then Z will have the same value at every point on the line segment joining those two corner points. This happens because $Z=ax+by$ is a linear function. So, if it is maximum at two endpoints of an edge of the feasible region, then it remains maximum at all points on that edge. Hence, $Z_{\text{(max)}}$ occurs at infinitely many points. So, the correct option is: Option D: Infinity

Benchmark Problem 2: KCET 2026 (Q#2)

The corner points of the feasible region determined by the system of linear constraints are $(0,10), (5,5), (15,15), (0,20)$. Let $z=px+qy$ where $p,q>0$. The relation between p and q , so that the maximum z occurs at both points $(15,15)$ and $(0,20)$ is

- (A) $p=q$
- (B) $p=2q$
- (C) $q=2p$
- (D) $q=3p$ [Correct]

Correct Answer: D

Solution Analysis: For the maximum value of $z=px+qy$ to occur at both points $(15,15)$ and $(0,20)$, the value of z at these two points must be equal. At $(15,15)$, $z=15p+15q$ At $(0,20)$, $z=0p+20q=20q$ Since maximum occurs at both points, $15p+15q=20q$ $15p=5q$ $q=3p$ So, the required relation is $q=3p$ Hence, the correct option is Option D.

Benchmark Problem 3: KCET 2022 (Q#4)

A dietician has to develop a special diet using two foods X and Y. Each packet (containing 30g) of food X contains 12 units of calcium, 4 units of iron, 6 units of cholesterol and 6 units of vitamin A. Each packet of the same quantity of food Y contains 3 units of calcium, 20 units of iron, 4 units of cholesterol and 3 units of vitamin A. The diet requires at least 240 units of calcium, at least 460 units of iron and at most 300 units of cholesterol. The corner points of the feasible region are

- (A) $(2,72), (40,15), (15,20)$
- (B) $(2,72), (15,20), (0,23)$
- (C) $(0,23), (40,15), (2,72)$
- (D) $(2,72), (40,15), (115,0)$

Correct Answer: None

Solution Analysis: Let x and y be the number of packets of food X and Y $x \geq 0, y \geq 0 \Rightarrow 12x+3y \geq 240 \Rightarrow 4x+y \geq 80$ (i) $4x+20y \geq 460 \Rightarrow x+5y \geq 115$ (ii) $6x+4y \leq 300 \Rightarrow 3x+2y \leq 150$ (iii) $\Rightarrow x \geq 0, y \geq 0$ (iv) $4x+y \geq 80 \Rightarrow x \geq 20$ $y \geq 80-4x$ $x+5y \geq 115 \Rightarrow x \geq 115-5y$ $3x+2y \leq 150 \Rightarrow x \leq 50-2y$ $0 \leq 50-2y \Rightarrow y \leq 25$ $0 \leq 80-4x \Rightarrow x \leq 20$ Intersection points of the lines are shown in the graph we can see that corner point of the feasible region A, B and C